

# CONTINUATION — HelixTranslate session-resumption file

**Revision:** 5 **Last modified:** 2026-06-13T13:30:00Z **Purpose:** Single canonical out-of-the-box entry point for any fresh session (§11.4.131 / §12.10 / §11.4.127). To resume: point a new session at THIS file, run `git fetch --all`, and say **continue**.

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## SHORT resumption sentence

Read docs/CONTINUATION.md then `.remember/remember.md`, run `git fetch --all`, and **continue** the HelixTranslate “raise-to-enterprise” mandate at **Phase 4 (tests/Challenges/HelixQA to ~100% per type + expand real ebook-translation evidence)** while finishing the Phase-3 tail (Models decision, governance gaps). Binding: anti-bluff §11.4 (real captured evidence, no false PASS), no-force-push §11.4.113, flat snake\_case submodules §11.4.28/§11.4.29, every change independently reviewed §11.4.142, containers-first infra §11.4.76, release tags prefixed helix\_translate- (§11.4.151). Work subagent-driven (§11.4.70) with  $\geq 3$  parallel streams (§11.4.103).

## Live state anchors (moment-valid)

- **Parent HEAD:** 7331c54 (wave2: D2 race fix + W2-A/W2-B coverage; prior e775088 batch W4/W5/W6/W8/W9/W10/W12 + 8a6af67 doc-sync) on main; pushed to BOTH milos85vasic/Translator + HelixDevelopment/HelixTranslate (verified at 7331c54, fast-forward, no force).
- **llm\_provider HEAD:** b3dc7f9 (W7 CONST-036 propagation block + regenerated html/pdf), pushed to HelixDevelopment/LLMProvider master.
- **Constitution HEAD:** 5e671fe (§11.4.151 added), pushed to all 6 upstreams; parent pointer bumped.
- **Build:** `go build ./... = EXIT 0`. **Total test coverage = 50.7%** (`go tool cover -func total`; see docs/testing/coverage\_matrix.md).
- **Submodules (flat, snake\_case):** challenges containers constitution doc\_processor docs\_chain helix\_qa llm\_orchestrator llm\_provider llms\_verifier security vision\_engine.
- **Release prefix:** HELIX\_RELEASE\_PREFIX=helix\_translate (.env git-ignored, .env.example tracked). First RC: helix\_translate-1.0.0-dev-0.0.1 — create ONLY when genuinely green (§11.4.40), operator-gated.
- **Safety backup:** ../helix\_translate\_gitbackup\_20260611T181545Z/repo.git.mirror (pre-rename 311c585).

# DONE

**Phase 1 — flat snake\_case submodules** (reviewed GO, pushed): 8 submodules renamed; Go pkg challenges/→pkg/challenge\_runner/; docs\_chain added; .env release-prefix.go build ./... EXIT 0. **Phase 2 — constitution §11.4.151** release-prefix rule → pushed to all 6 constitution upstreams; parent pointer bumped; inheritance meta-test PASS. **Phase 3 (partial):** - Security/ plain dir → flat security submodule (proven byte-identical to upstream, zero loss; §11.4.124). - All submodules + main synced to every upstream (one stale containers gitlab mirror FF'd). - **docs\_chain OPERATIONAL:** .docs\_chain/contexts/tracked\_docs.yaml registered; real pandoc+weasyprint sync produced docs/CONTINUATION.{html,pdf} + README.{html,pdf}; verify = in-sync. See docs/DOCS\_CHAIN.md. - **Real translation PROVEN (anti-bluff):** EN→Serbian via DeepSeek deepseek-chat, 208 Cyrillic chars, differs from source, no placeholders. Evidence: qa-results/translation/<ts>/ (raw, git-ignored). Asset: test/assets/crow\_and\_pitcher\_en.txt. - **Fixed a regression I introduced:** challenge scripts' PROJECT\_ROOT="\${SCRIPT\_DIR}/../.." broke when moved 1 level deeper → corrected to ../../..; 3 challenges re-verified PASS (cache\_invalidation, model\_verification\_gate, anti\_bluff step1-3).

## CLOSED this session (2026-06-13, batch e775088 / llm\_provider b3dc7f9 — all pushed, captured evidence)

- **W4** pkg/security flaky Short-TTL — widened TTL→1min; 50× + -race + CPU-load all PASS.
- **W5** pkg/distributed 120s timeout — bounded SMTP send (real defect) + in-process test servers; 3× ~48.5s, vet 0.
- **W6** internal/verifier/discovery 600s timeout — injectable URLs (prod unchanged) + sentinel stub + honest SKIP; 3× ~1s, paired §1.1 mutation proven.
- **W7** llm\_provider CONST-036 propagation block — challenge GREEN; html/pdf regenerated.
- **W8** anti\_bluff\_execution\_challenge.sh BSD-sed — portable tmpfile sed (§11.4.67); isolated rc=0 proof.
- **W9** tracked .bak/.backup residue — git-history-investigated (§11.4.124), removed + .gitignore; build 0.
- **W10** -script latin→Cyrillic — pkg/script was never imported into unified-translator; normalizeScript wired; RED→GREEN 7 tests + 3×.
- **W12** 8 challenge scripts brittle root — git -C rev-parse --show-toplevel w/ fallback (§11.4.111); all parse:OK root:OK.
- **D2** (wave2, 7331c54) test/unit intermittent FAIL — TestVerifier/EventEmission timing race (50ms Sleep vs goroutine-dispatched handler) + data race (proven -race). Mutex eventCollector + bounded poll. BEFORE 1/10 FAIL+race → AFTER -race -count=10 10/10 PASS.
- **W2-A** (wave2) cmd/unified-translator coverage 3.2%→18.0%, mutation-verified; + latent dangling-pointer fix (Steps []TranslationStep→[]\*TranslationStep).
- **W2-B** (wave2) pkg/storage coverage 32.9%→35.3% (Redis cache-key helpers 0%→100%, real SQLite round-trips), mutation-verified, daemon paths honest-SKIP.

- **W2 (wave3, in-flight)**: pkg/translator stress/chaos (W3), pkg/sshworker, pkg/batch coverage — subagents running.

## OPEN workable items (queue — many parallelizable, disjoint scope)

| #   | Item                                                                          | Type                               | Notes / evidence                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----|-------------------------------------------------------------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| W1  | Models/ divergent types.go (60-line lean LLMRequest, absent upstream)         | <b>Operator decision §11.4.122</b> | Convert needs preserving/ discarding the local shape. Not consumed (no replace/import). DO NOT silently remove. Deferred — not blocking; surface to operator before W16.<br><br>Done: unified-translator 3.2%→18%, storage 32.9%→35.3%. In-flight (wave3): sshworker(19.6%), batch. NOTE: many 0% pkgs (gRPC/api-server/redis/postgres) are integration-infragated — unit gains capped; honest deferral per package. Core translator 77–82%. |
| W2  | Coverage → ~100% (§11.4.27); ongoing                                          | Task                               |                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| W3  | Missing test types: chaos, ddos, scaling, ui, ux, full-automation (in-module) | Task                               | §11.4.85 chaos/ stress mandatory.                                                                                                                                                                                                                                                                                                                                                                                                            |
| W11 | Wire more docs_chain contexts (Issues/Fixed/Status once they exist)           | Task                               | docs_chain proven operational.                                                                                                                                                                                                                                                                                                                                                                                                               |
| W13 | .gitmodules section labels still PascalCase                                   | Task (cosmetic)                    | Needs per-submodule .git/modules/* gitdir move.                                                                                                                                                                                                                                                                                                                                                                                              |
| W14 | No project commit wrapper scripts/                                            | Task                               |                                                                                                                                                                                                                                                                                                                                                                                                                                              |

| #   | Item                                                                                  | Type    | Notes / evidence                                                                                                                                                                                                            |
|-----|---------------------------------------------------------------------------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | commit_all.sh<br>(§11.4.22)                                                           |         | Multi-upstream<br>push + sibling-doc<br>checks (§11.4.75).                                                                                                                                                                  |
| W15 | Containers-first infra<br>(§11.4.76) — run<br>services via<br>containers<br>submodule | Feature | Phase 3/5.                                                                                                                                                                                                                  |
| W16 | Convert Models/<br>→models submodule<br>after W1 resolved                             | Task    | Blocked on W1.                                                                                                                                                                                                              |
| D1  | docs_chain +<br>security<br>CONSTITUTION.md<br>lack literal<br>CONST-036              | Task    | §11.4.118<br>discovery. They<br>use §11.4.X<br>cascade scheme<br>(not CONST-<br>NNN); challenge<br>doesn't check<br>them.<br>Governance-<br>design decision<br>needed — do<br>NOT blindly inject<br>CONST-NNN<br>(§11.4.6). |

## NEXT phases (priority order)

- **Phase 4** — drive W2–W8 + W10: per-type test suites to ~100% (§11.4.27), real Challenges + HelixQA bank execution with captured evidence, expand real ebook translations (all formats, whole chapters, anti-bluff §11.4/§11.4.69). Subagent-driven, ≥3 parallel streams.
- **Phase 5** — containers-first infra (W15); docs\_chain context expansion (W11); full docs (guides/diagrams/SQL/templates) + enterprise Website.
- **Phase 6** — RC tag helix\_translate-1.0.0-dev-0.0.1 across all submodules + main ONLY when genuinely green (§11.4.40), operator-confirmed.

## Binding constraints

Anti-bluff §11.4 (real captured evidence, no false PASS) · no-force-push §11.4.113 (merge-onto-latest-main) · multi-upstream push §2.1 · flat snake\_case §11.4.28/§11.4.29 · every change independently reviewed §11.4.142 · host-safety §12 (no suspend/poweroff) · no silent component removal §11.4.122 · containers-first §11.4.76 · release prefix §11.4.151 · subagent-driven default §11.4.70 · ≥3 parallel streams §11.4.103.

## Gotchas

- `/Volumes/T7` is **case-sensitive HFS+** (NOT the internal disk — verify on the actual volume, §11.4.6).
- No `scripts/commit_all.sh`; stage explicit paths, never `git add -A`; never `git add helix_qa` (nested third-party pointer drift).
- `PIPESTATUS` empty in this shell (zsh); use `$?` directly.
- Mutation challenges can leave `.go.backup/.bak` residue — scan + clean before commit (§11.4.84).